



TRANSGUARD AI

Data Intelligence • Predictive Analytics • Smart Maintenance

Enrollment No:	Name:
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Role of Team Members

1

Member 1 — Hardware & IoT Developer

Select sensors, connect the low-voltage transformer prototype, program ESP32, collect real-time sensor data, and handle hardware testing.

2

Member 2 — AI/ML Developer

Collect and preprocess datasets, select ML algorithms, train the failure-prediction model, evaluate accuracy, and generate the risk score.

3

Member 3 — Backend & Database Developer

Develop APIs using Node.js/Python, store sensor readings and alerts, connect the AI model with the backend, and manage database operations.

4

Member 4 — Frontend & Documentation Developer

Design the React dashboard, display sensor values and health score, show alerts and predictions, prepare diagrams, documentation, and presentation.

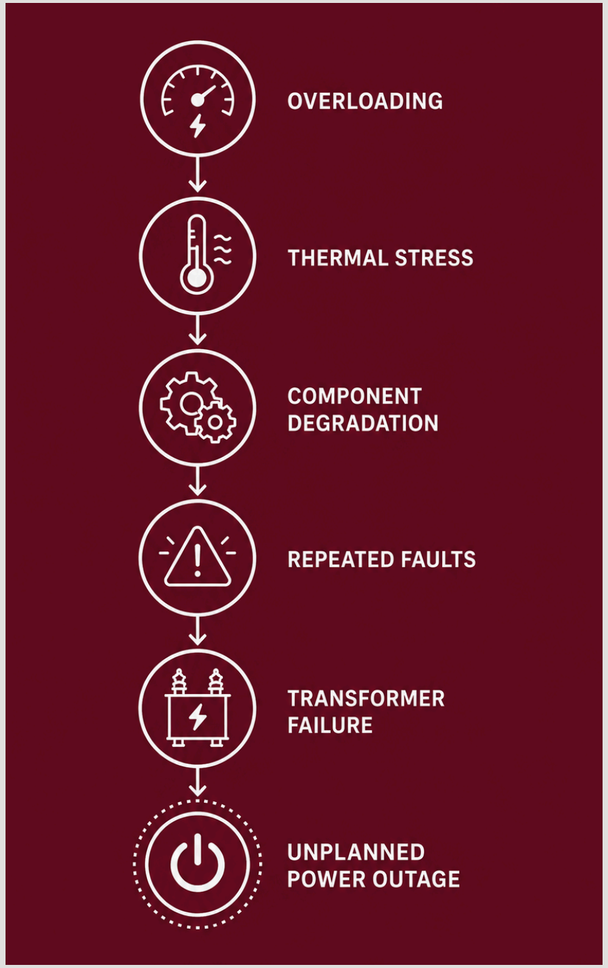
Abstract Of Definition

AI-based smart transformer failure prediction is a predictive maintenance approach that analyzes transformer operating and historical data to assess asset health, identify abnormal patterns, estimate failure risk, and recommend timely maintenance before a critical failure occurs.

From Failure Response to Failure Foresight

Our system does not wait for a transformer to fail.It uses operational data, historical behaviour , AI/ML to identify early warning patterns and estimate the transformer's failure risk.

Problems solved by our idea



01 **Hidden Risk**

Identifies risks that may not be obvious from a single parameter.

02 **Delayed Detection**

Detects combinations of abnormal operating conditions earlier.

03 **Maintenance Uncertainty**

Helps prioritize which transformers require attention first.

04 **Repeated Failures**

Uses historical fault and maintenance information to identify recurring patterns.

05 **Repeated Failures**

Supports a transition toward condition-aware, predictive maintenance.

Market Landscape

TXpert™ | Hitachi Energy

Focus: Digital transformer monitoring & asset health

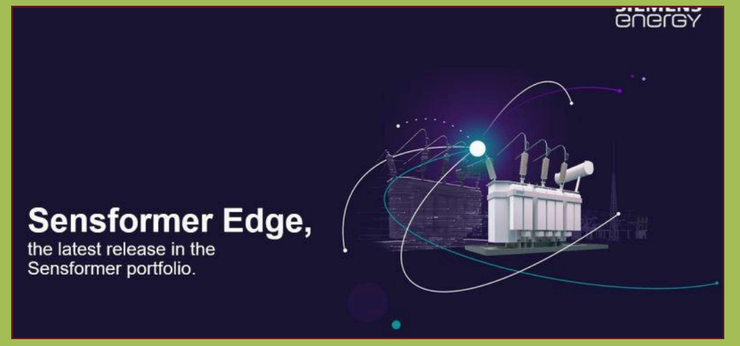
- Condition & Trend monitoring
- Operational data analysis
- Early-warning capabilities
- Condition-based maintenance



Sensformer® | Siemens Energy

Focus: Digitalization of transformer assets

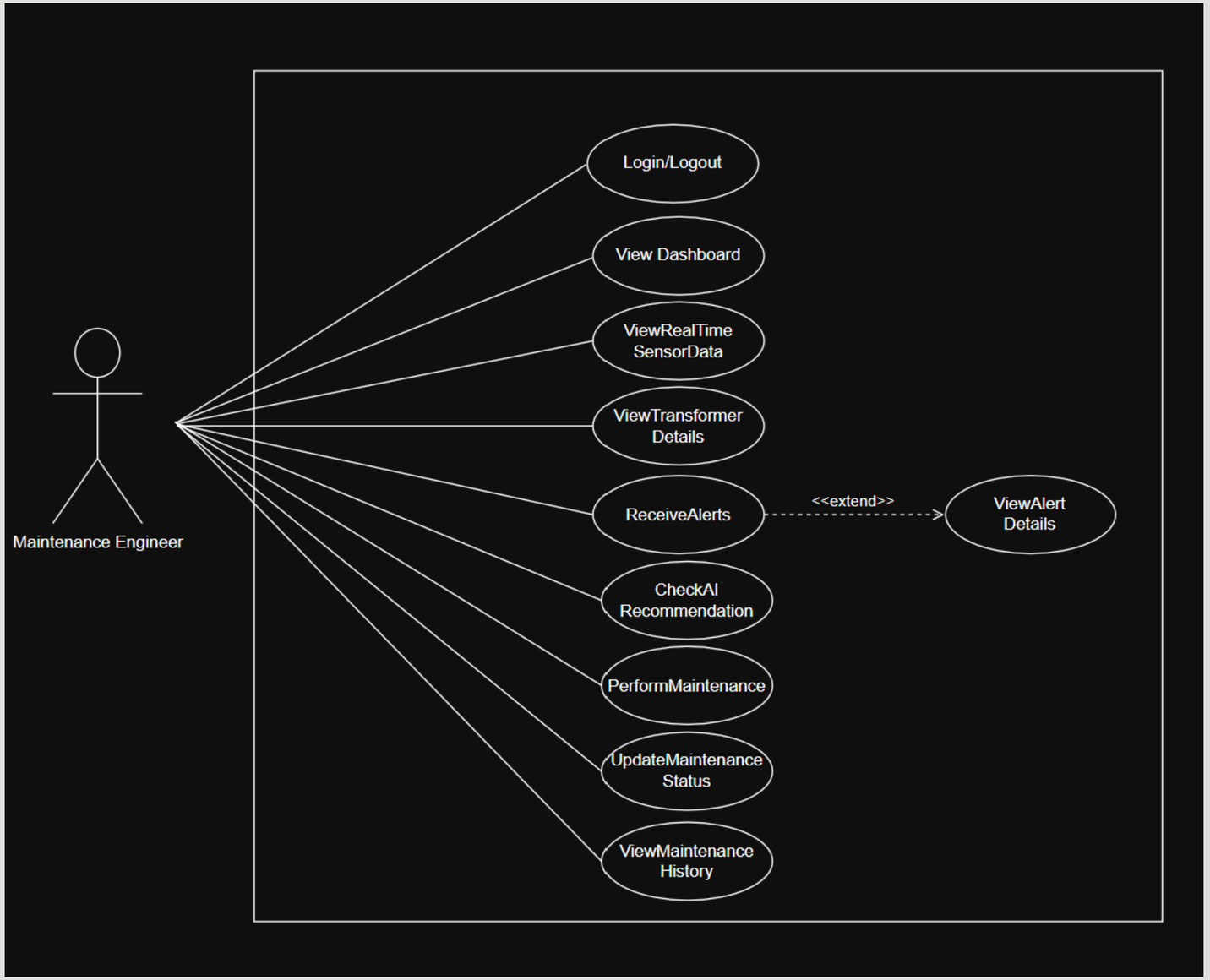
- Operational monitoring
- Transformer data collection
- Asset condition insights
- Predictive/analytical capabilities



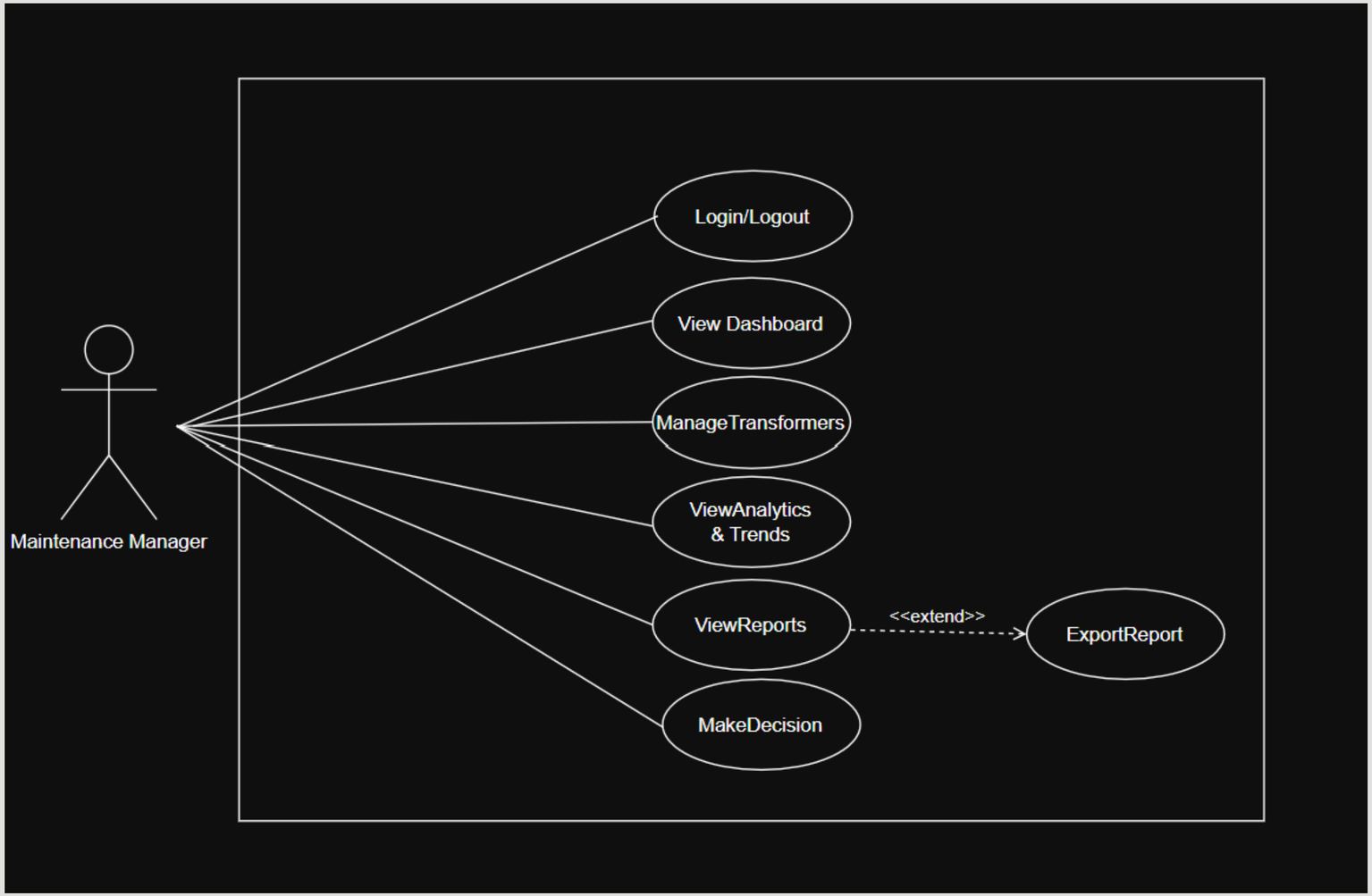
Feature Comparison: Transguard AI vs. Existing Apps

Features	Our Proposed System	TXpert™	Sensformer®
Temperature Monitoring	★	✓	✓
Load Analysis	★	✓	✓
Ageing Assessment	★	✓	✓
Fault History Analysis	★	◐	✓
AI/ML Failure Prediction	★	◐	◐
Health Score	★	✓	✓
Failure Risk Estimation	★	✓	◐
Maintenance Priority Score	★	✓	✓
Simple LOW / MEDIUM / HIGH Risk Output	★	-	-
Fleet-wise Risk Ranking	★	✓	✓

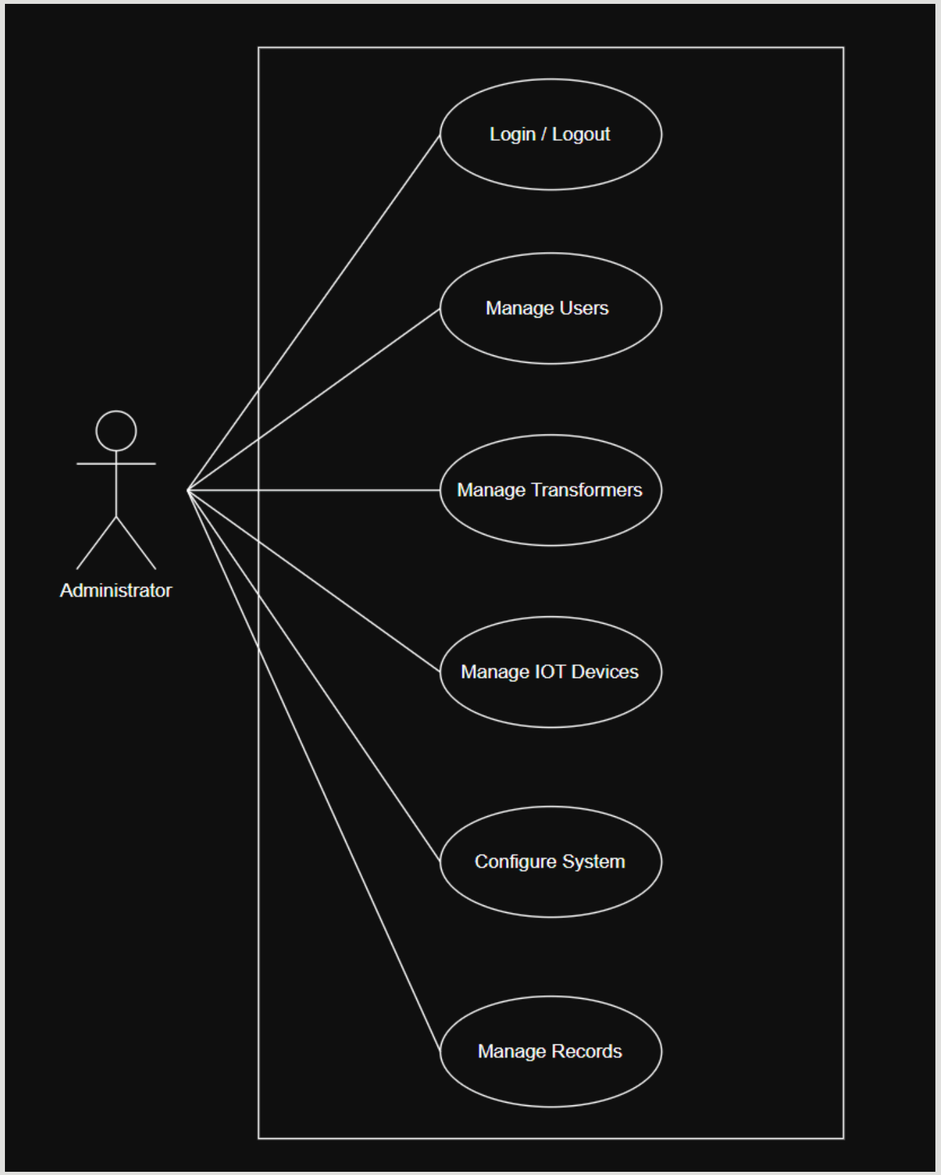
Maintenance Engineer- Use Case Diagram



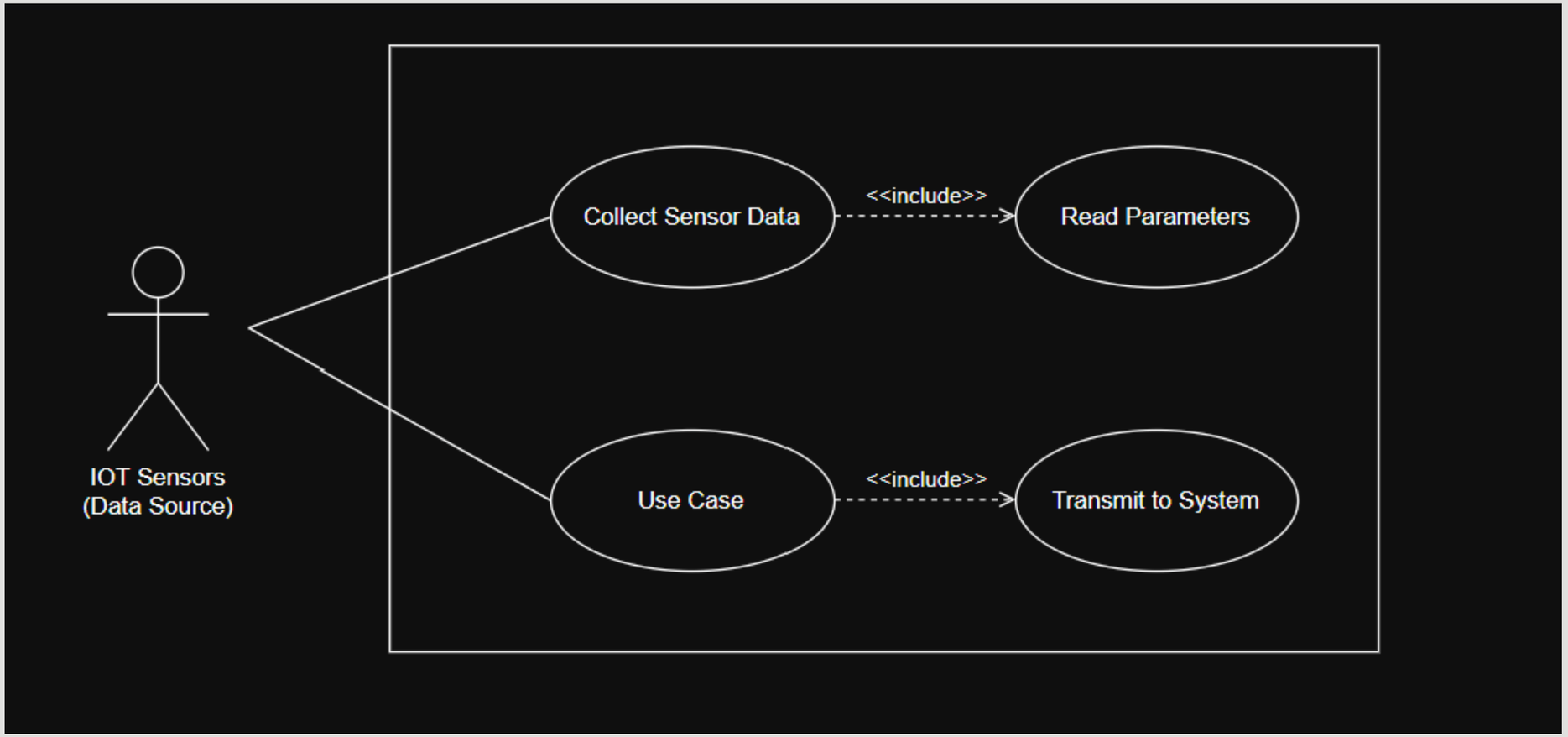
Maintenance Manager- Use Case Diagram



Administrator - Use Case Diagram



IOT Sensors - Use Case Diagram



Future Directions

Real-Time Monitoring

Integrated IoT sensors to continuously collect temperature, load, voltage, current and other transformer parameters.

Advanced Prediction

Improve AI/ML models using larger historical and multi-parameter datasets

Failure Classification

Extend the system to identify the possible type of transformer fault, not only the overall failure risk.

Automated Maintenance Planning

Generate maintenance schedules based on predicted risk and priority

Fleet Intelligence

Monitor multiple transformers together and rank them according to their health score and failure risk.

Real-Time Alerts & Notifications

Send immediate notifications to maintenance engineers when abnormal conditions or high-risk patterns are detected.

References

- Hitachi Energy – TXpert™ — Digital transformer monitoring and asset-health management.
- Siemens Energy – Sensformer® — Digital transformer monitoring, data collection and asset-condition insights.
- Transformer Predictive Maintenance Research — AI/ML-based approaches for transformer health and failure prediction.
- IoT-Based Transformer Monitoring Studies — Sensor-based monitoring of transformer operating conditions.

thank you